1 Introduction

Common environments are available: `definition`, `theorem`, `lemma`, `remark`, `corollary`, `problem`, `claim`.

Definition 1. A metric space (X, d) is *complete* if every Cauchy sequence converges in X.

Theorem 1. Every compact metric space is complete.

Remark 1. The converse is not true: $\mathbb{R}$ is complete but not compact.

Lemma 1. Let (X, d) be compact. For any $\varepsilon > 0$, finitely many ε-balls cover X.

Corollary 1. A closed bounded subset of $\mathbb{R}^n$ is compact.

Problem 1. Prove that $[0, 1]$ with the standard metric is complete.

2 More Content

This template supports `\todo` notes and math-heavy content. Use `claim`, `proposition`, and other environments as needed.

Example todo note.

The following identity is useful:

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}.$$